



Autonomous Vehicles: Navigating a Driver-less World

This presentation explores the legal and regulatory landscape of autonomous vehicles. We'll cover the physical ecosystem, strategic partnerships, and key legal issues. This includes regulatory frameworks, product liability, cybersecurity, and intellectual property. We will examine a case study of AVs in public transportation.





Agenda: Key Areas of Focus

1 SAE Levels & Ecosystem

Understanding the levels of automation and the vehicle's physical components.

2 Legal Issues & Regulations

Navigating product liability, data privacy, and intellectual property rights.

3 Strategic Partnerships

Examining collaborations that drive innovation and deployment of AV technology.

4 Case Study

Analyzing Keolis and the implementation of AVs in public transport.



Introduction to Autonomous Vehicles

Autonomous vehicles are poised to revolutionize transportation. They promise increased safety, efficiency, and accessibility. However, their deployment raises complex legal and regulatory challenges. This presentation will navigate these challenges.



Basic Physical Ecosystem



GPS & LIDAR

Global Positioning provides location. Light detection creates 3D maps.

Cameras & Sensors

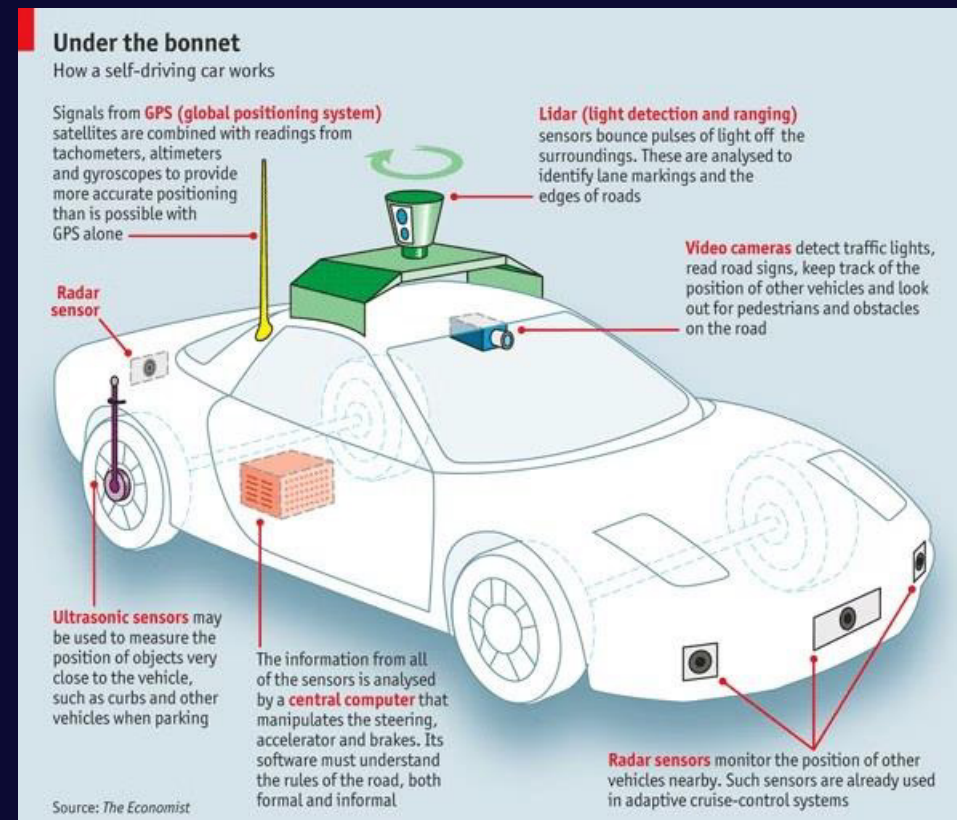
Cameras enable video.
Ultrasonic sensors measure distance.

Central Computer

The vehicle's brain receives info and directs operation.

Radar

Radio waves are key for short and long-range depth detection.



Key Components: AV Technology

1

Cameras

Real-time obstacle and lane detection. Tracks roadway info and road signs.

2

Radar & LIDAR

Radar detects short and long range depth. LIDAR creates 3D maps.

3

GPS & Sensors

GPS triangulates car position. Ultrasonic sensors calculate short-range distance.

4

Central Computer

Brain of the vehicle. Receives information and directs operation.





Strategic Partnerships in AV Development

Collaboration is key to advancing autonomous vehicle technology. Automakers, tech companies, and suppliers are forming strategic alliances. These partnerships accelerate innovation and deployment.



Intel



Nvidia



Waymo

Federal Regulation Overview

1

DOT Guidance (2017)

ADS 2.0: A Vision for Safety outlines key safety elements for autonomous vehicles.

2

Safety Elements

Includes system safety, operational design, and object detection and response.

3

Voluntary Self-Assessment

Entities involved in ADS testing should publish a Voluntary Safety Self-Assessment.



Managing Liability Among Parties

Scope & Responsibility

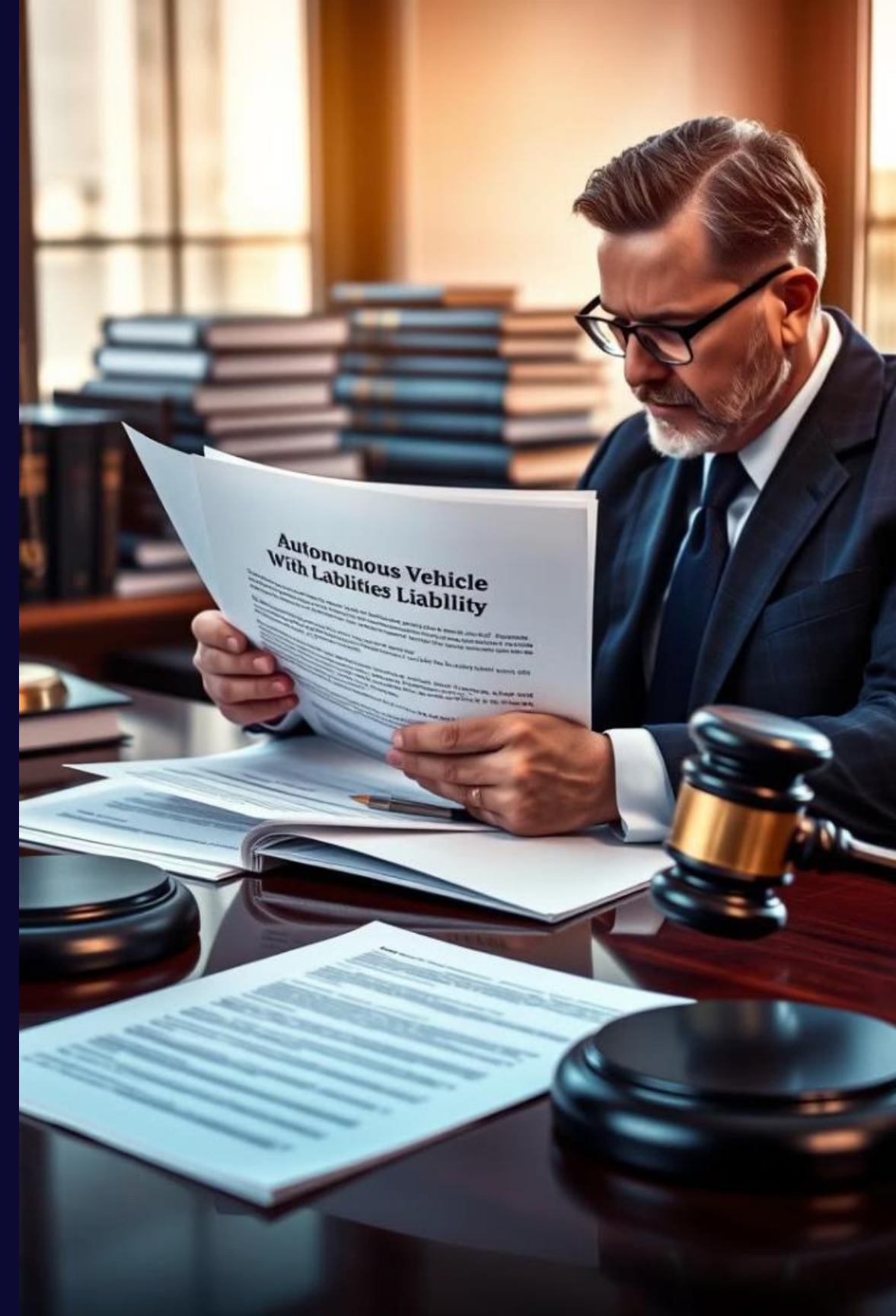
Warranties and indemnifications should clearly define liability.

Maintenance & Updates

Responsibility for repairs and updates needs to be defined.

Liability

Define liability between automaker, tech company, and owner.



Common Security Vulnerabilities



Code volume increases chances for exploitation. Lack of source control is a weakness. Apps interfacing with cars could cause spread of malware.

Cybersecurity Takeaways

1. Provide multi-layered protection across all levels of the vehicle's systems.
2. Defend against potential gateways, especially externally-facing mechanisms.
3. Ensure vendors and suppliers maintain strong security practices.
4. Promote the delivery of timely updates to address vulnerabilities.

